Credit Card Customer and Transaction Analytics Dashboard

Python data preparation and Power BI dashboard

1. Business Problem

The bank issues credit cards but does not have one clear view of how customers use them. It does not know which customers spend the most, how they pay back their bills, how much of their credit limit they use, or which customers may become a credit risk. Without this view it is hard to grow card usage and to control losses.

2. Business Objectives

- Measure the total number and value of card transactions.
- Find the customer segments and merchant categories that spend the most.
- Track monthly and yearly trends in spending and payments.
- Study credit limits, credit utilization and outstanding balances.
- Understand payment behavior and find high risk customers.
- Find high value customers and customers who can be targeted to use their card more.

3. Dataset

Source: Kaggle dataset "computingvictor/transactions-fraud-datasets". It holds data of a US bank from January 2010 to October 2019. We used four files:

File	What it contains	Rows
users_data.csv	Customer age, gender, income, debt, credit score	2,000
cards_data.csv	Card brand, card type, credit limit	6,146
transactions_data.csv	Date, amount, merchant, city, state, category	13.3 million
mcc_codes.json	Merchant category names	109

This project studies credit cards only. After cleaning we kept 855 customers, 1,351 credit cards and 4.1 million transactions.

4. Data Cleaning and Preparation

All cleaning was done in Python (pandas) in the notebook CreditCard_Analytics.ipynb.

Table	What we did
Customers	Removed $ signs from income and debt and changed them to numbers
Customers	Replaced 8 unrealistic incomes (below $1,000) with the median income
Customers	Removed address and birth details (personal data, not needed)
Cards	Kept only credit cards (2,057 of 6,146 cards)
Cards	Replaced 26 credit limits of $0 with the median limit ($10,100)
Cards	Removed card number and CVV (sensitive data)
Transactions	Read the 1.2 GB file in pieces of 1 million rows
Transactions	Removed $ signs and changed date text into real dates
Transactions	Filled empty states with Online and added a location type
Transactions	61,828 failed transactions and 216,832 refunds are not counted as spending
Checks	No duplicate transactions and no missing values after cleaning

New columns we created

- Customer groups: age group, income segment, credit score band and debt to income band.
- Monthly summary for every card: transactions, spending, refunds and declined transactions.
- Payments and balances (estimated, see the note below).
- Credit utilization: the bill divided by the credit limit, for the last 12 months.
- Risk score (0 to 100) from credit score, debt to income, utilization and declined payments.
- Value segment (top 10% spenders are High Value) and an action group for every customer.

Note about payments

The dataset has no payment records. So we estimated payments with a simple rule that uses each customer's real credit score and debt to income. Full Payers (score 740 or more) pay 100% of the bill, Partial Payers pay 40% and Minimum Payers (score below 670 or very high debt) pay 10%. Payments, outstanding balances and utilization in this report are based on this rule.

Data model in Power BI

The clean data was saved in one Excel file (CreditCard_Clean_Data.xlsx) as a star schema: 5 dimension tables (customer, card, merchant category, year, date) and 3 fact tables (card by month, category by year, location by year). The dashboard uses 38 DAX measures.

5. Dashboard

The Power BI dashboard (CreditCard_Dashboard.pbix) has 5 pages and 15 charts. Every chart answers one business question.

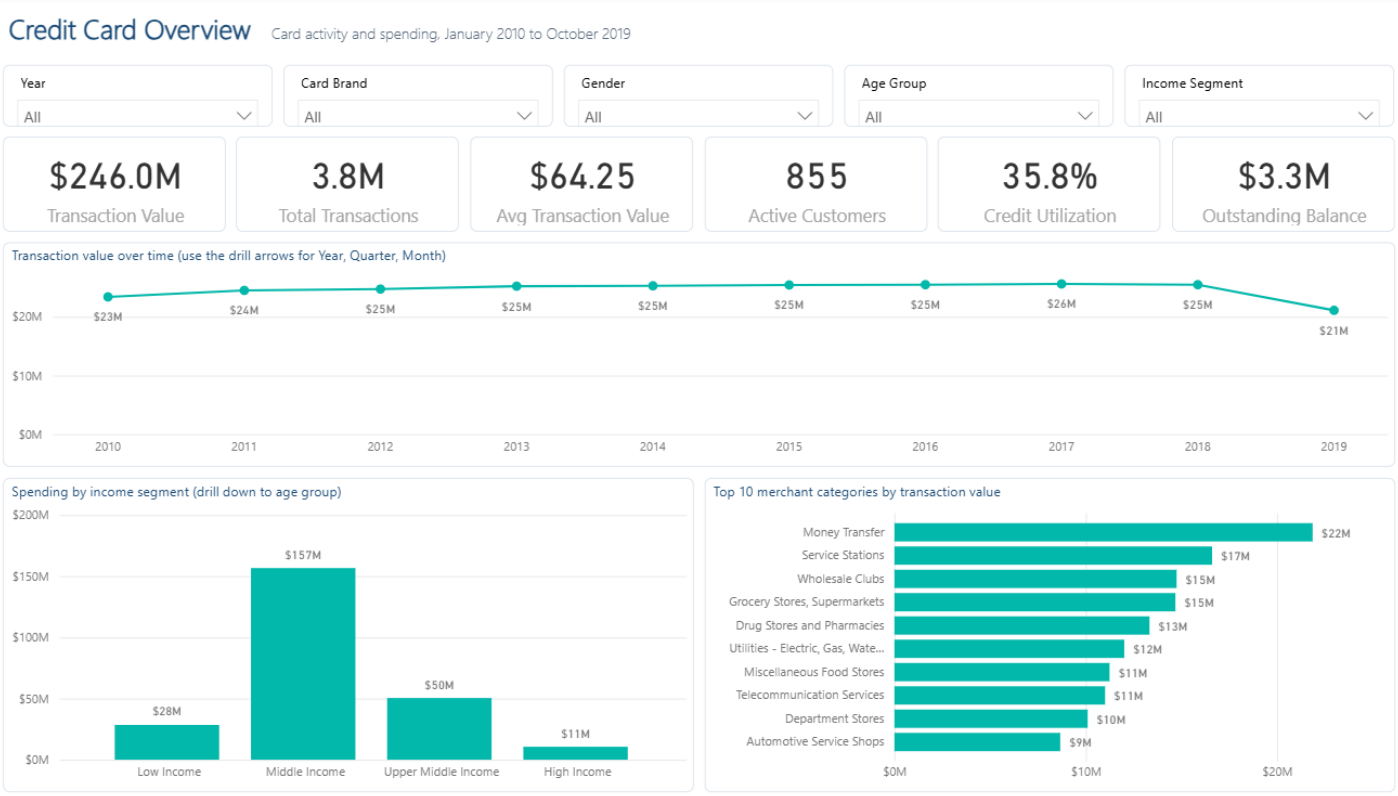
Page	What it shows
Overview	KPI cards, value trend, spending by segment, top categories
Merchants and Locations	Average value by category, top states, location type
Credit and Payments	Limit vs spending, utilization, payments, top balances
Risk and Opportunity	Risk by score band and age, action groups, high value profile
Customer Detail	One customer, opened by drill through

Dashboard features

- KPI cards: transaction value, transactions, average transaction, active customers, utilization, balance.
- Slicers on every page (synced): Year, Card Brand, Gender, Age Group, Income Segment.
- Drill down: Year to Quarter to Month on the trend charts, income segment to age group.
- Drill through: right click a customer to open the Customer Detail page.
- Tooltips: extra numbers appear when you hover over a chart.
- Conditional formatting: utilization and spending colors in the tables.
- Trend comparison: change vs last year and vs last month in the trend tooltips.

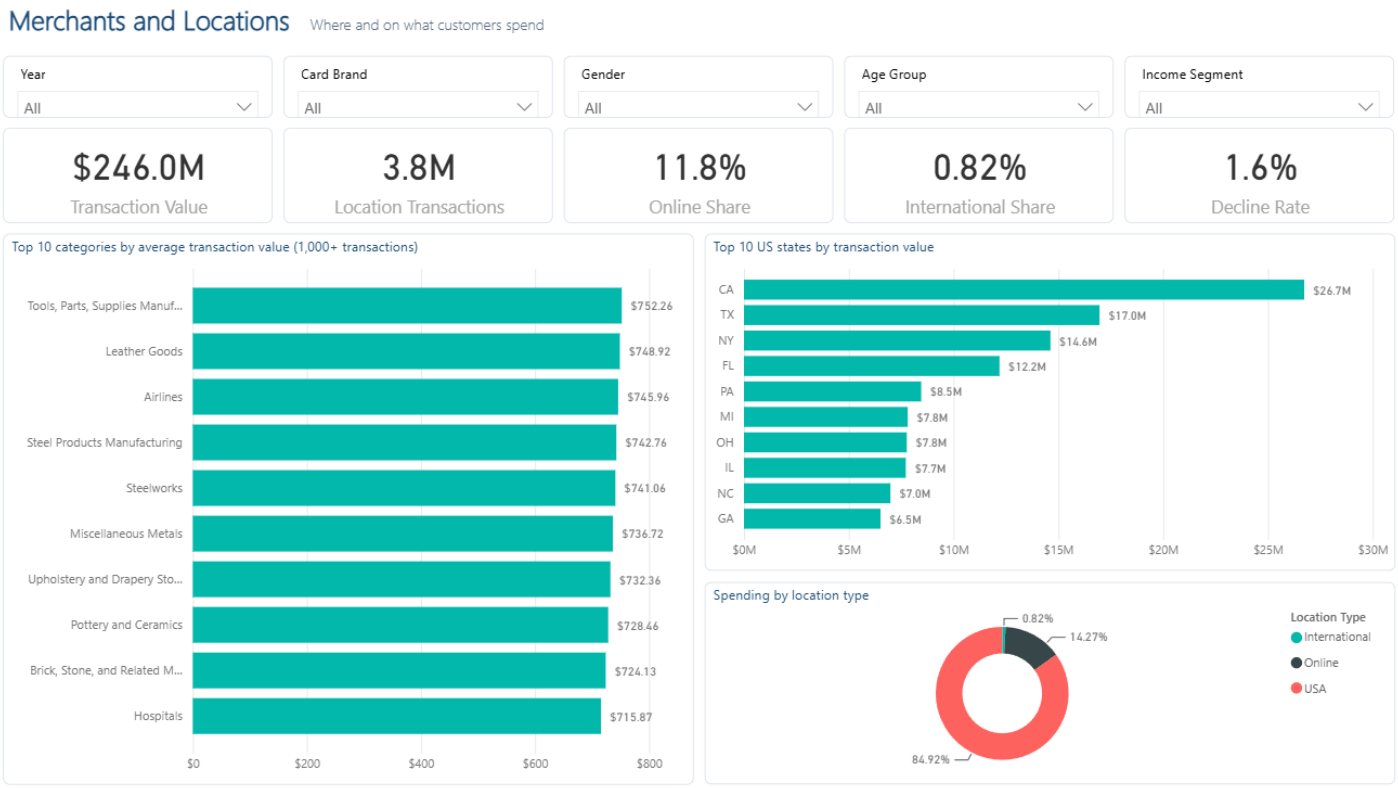
6. Analysis

6.1 Overview: transaction volume, value and trends



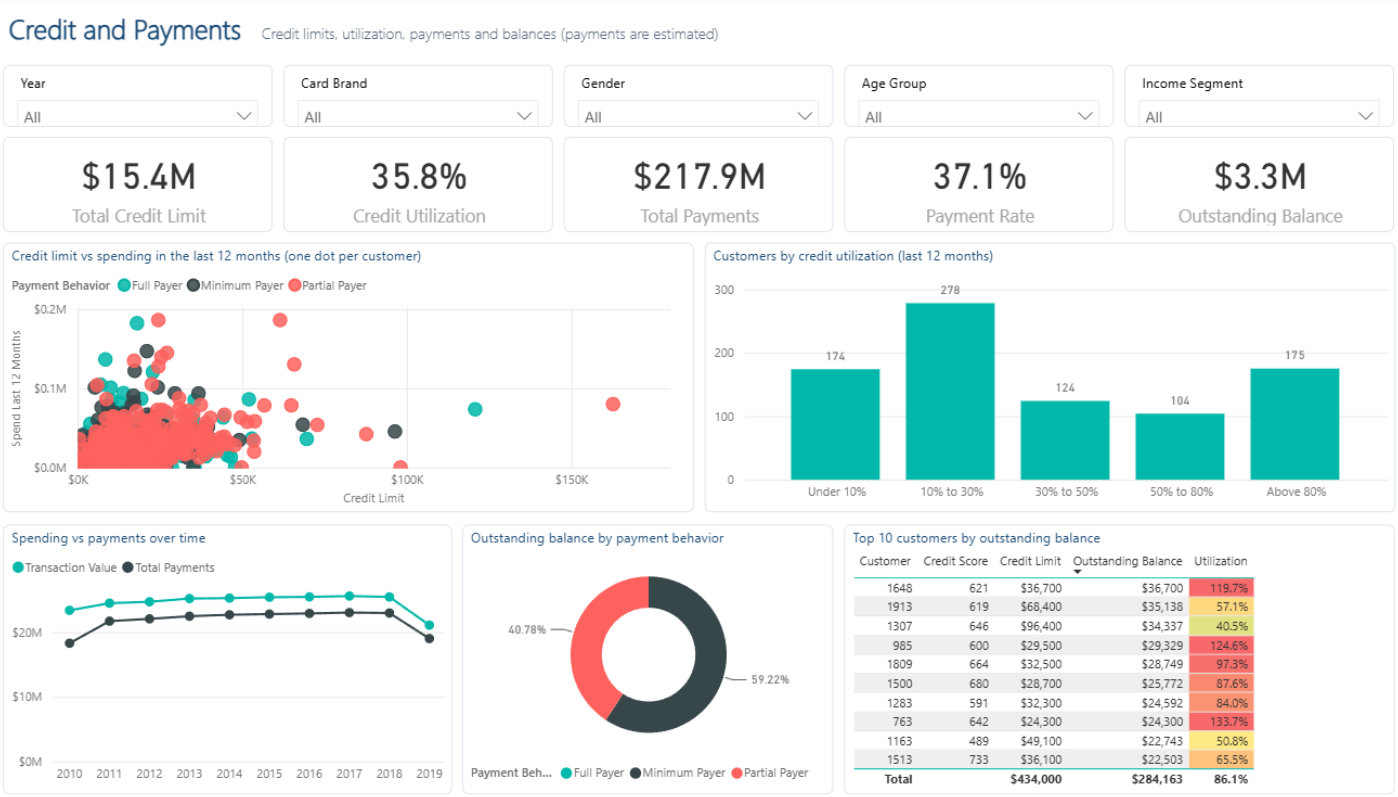
Customers made 3.8 million purchases worth $246.0 million, with an average of $64.25 per transaction. Spending is very steady at about $25 million a year from 2011 to 2018 (2019 has only 10 months). Month to month, spending moves only a little, with a small drop in February. Middle Income customers spend the most ($157 million, 64% of all spending).

6.2 Merchant categories and locations



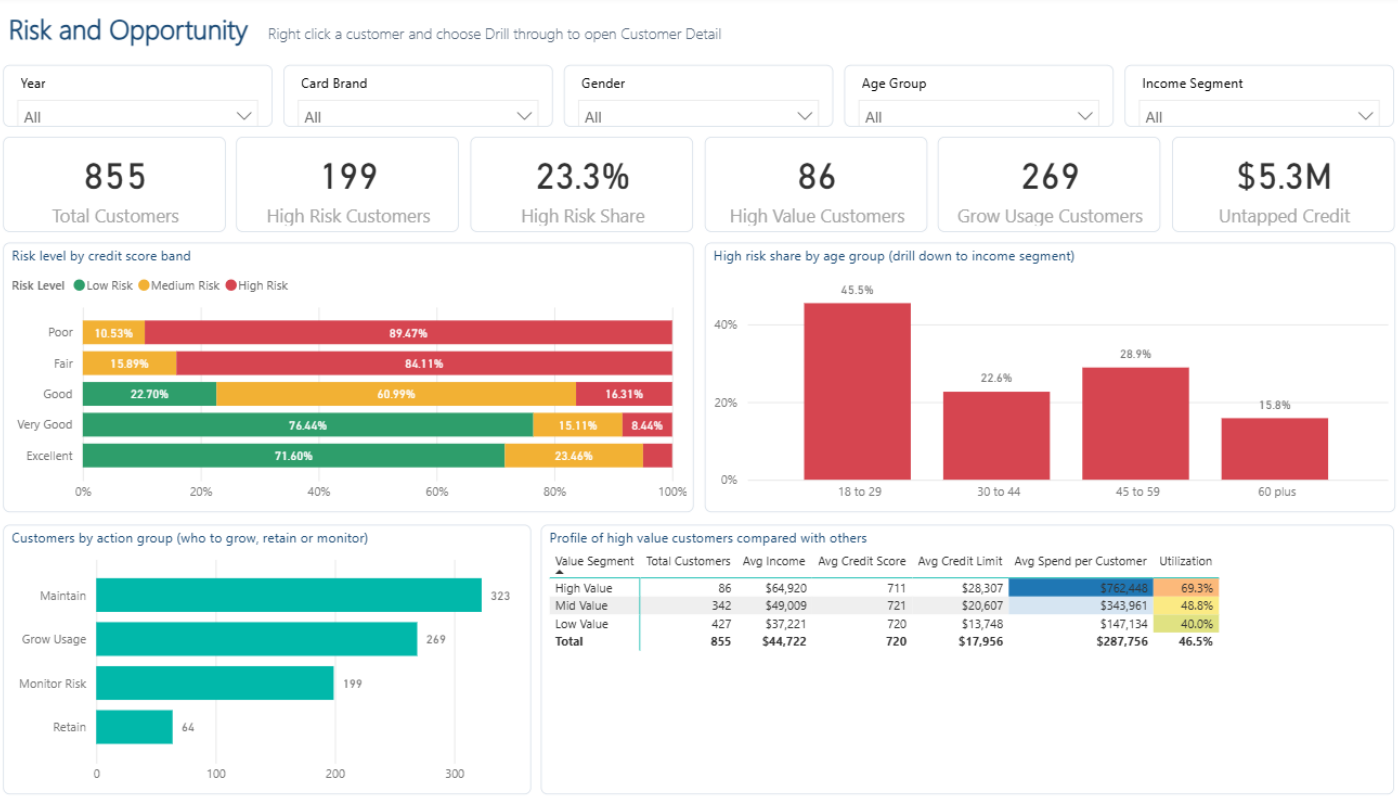
Money Transfer, Service Stations, Wholesale Clubs and Grocery Stores bring the highest value. The highest average transaction is in big ticket categories like tools, airlines and hospitals (about $700 or more). 85% of spending is in US shops, 14% online and 1% abroad. California, Texas, New York and Florida are the busiest states.

6.3 Credit limit, utilization, payments and balances



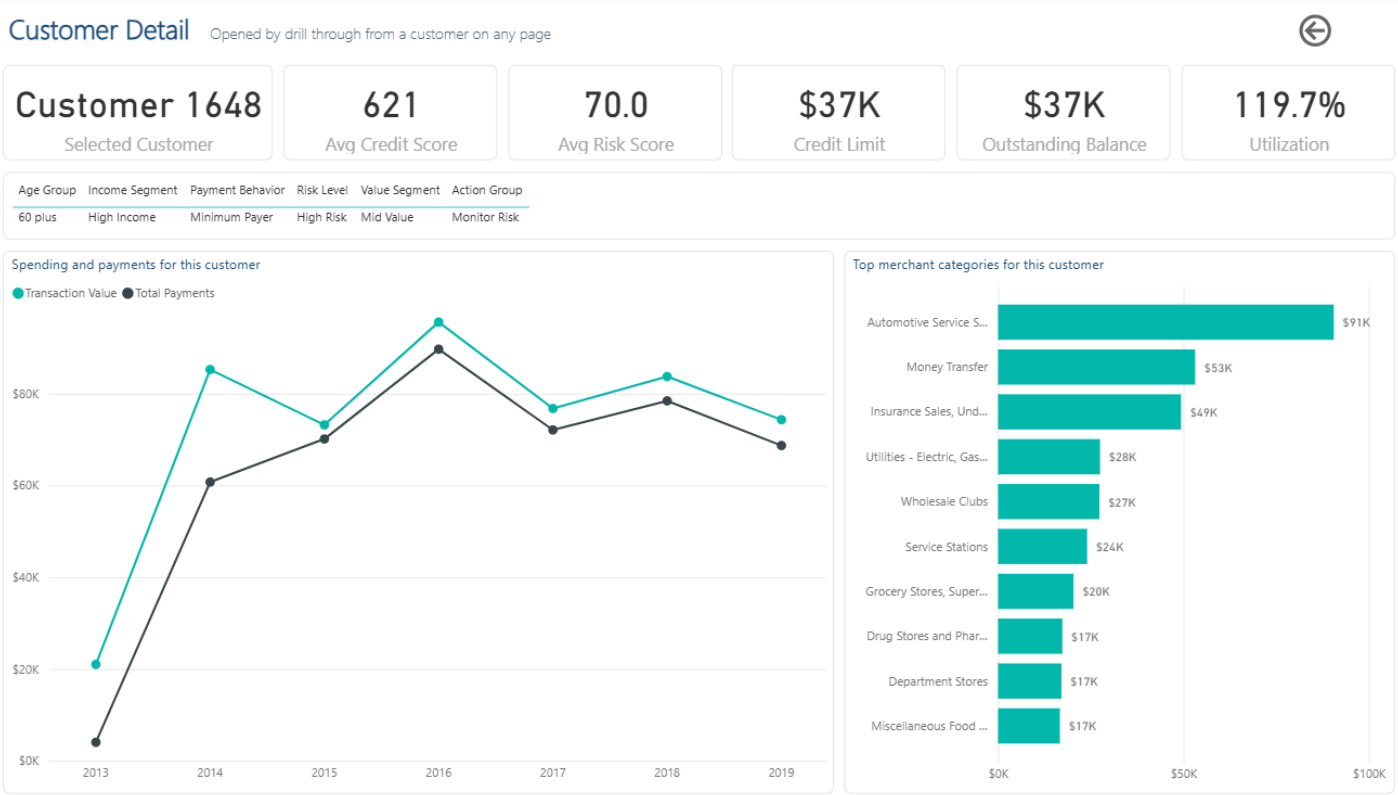
Total credit limit is $15.4 million and average utilization is 35.8%. Customers with higher limits spend more, but the link is weak (correlation 0.27). 175 customers use more than 80% of their limit. The outstanding balance is $3.3 million and Minimum Payers hold 59% of it. The top 10 balances all belong to Minimum Payers.

6.4 High risk and high value customers



199 customers (23%) are High Risk. Risk is highest for Poor and Fair credit scores (over 80% High Risk) and for young customers aged 18 to 29 (46%). High Value customers spend about 5 times more than Low Value customers, have higher income and higher limits, and use 69% of their limit. 269 low risk customers use less than 30% of their limit.

6.5 Customer Detail (drill through)



This page opens when you right click a customer and choose Drill through. The example is customer 1648: a Minimum Payer with a credit score of 621 who has used all of the $37K limit. The page shows the customer's spending and payments over time and where the customer spends.

7. Answers to the Business Questions

Question	Answer
1. Overall transaction volume and value?	3.8 million purchases worth $246.0 million, average $64.25.
2. Which segments spend the most?	Middle Income (64%) and customers aged 45 to 59 (39%).
3. Which categories have the highest value?	Money Transfer ($21.9M), Service Stations ($16.6M), Wholesale Clubs.
4. How does usage change month over month?	Very stable, about $2.1M a month. Small dip in February.
5. Who has the highest utilization?	175 customers above 80%, mostly with small limits and Minimum Payers.
6. Credit limit and spending?	Higher limits spend more, but the link is weak (0.27).
7. Who has big outstanding balances?	186 customers owe more than half their limit. All top 10 are Minimum Payers.
8. Main payment patterns?	247 Full Payers, 425 Partial Payers, 183 Minimum Payers (estimated).
9. Highest average transaction value?	Big ticket categories: tools, leather goods, airlines (about $750).
10. Which locations are most active?	US shops 85%. Top states: California, Texas, New York, Florida.
11. Which segments are risky?	Poor and Fair credit scores, age 18 to 29 and Minimum Payers.
12. Changes over time?	More cards each year, but spending per card fell from $23.3K to $19.0K.
13. Who to target for more usage?	269 low risk customers using under 30% of the limit ($5.3M unused).
14. Patterns of high value customers?	Higher income ($65K), higher limits ($28K), more cards, 69% utilization.

8. Executive Summary

Card spending is large and steady: $246 million over almost 10 years, about $25 million a year. Middle income customers and customers aged 45 to 59 bring most of the spending. Most money is spent on everyday needs such as money transfers, fuel, groceries and pharmacies, mostly in US shops.

The bank has two clear groups to act on. First, 199 High Risk customers, mostly with low credit scores who pay only a small part of their bill. They hold most of the $3.3 million outstanding balance. Second, 269 Low Risk customers who use less than 30% of their limit. They have $5.3 million of unused credit and are the best group to grow card usage. Spending per card has also gone down each year, so the bank needs to keep customers active.

9. Recommendations

- Grow usage: give the 269 Grow Usage customers cashback or rewards on fuel, groceries and online shopping to use more of their $5.3 million unused credit.
- Retain High Value customers: offer premium cards, higher limits and loyalty rewards to the 86 customers who spend the most.
- Control risk: watch the 199 High Risk customers closely, send payment reminders, and do not raise their limits until their payments improve.
- Help Minimum Payers: offer payment plans or balance transfer options to lower their balances.
- Be careful with young customers: start customers aged 18 to 29 with lower limits and teach them how to use credit well.
- Stop the drop in spending per card: run campaigns for cards that are used less each year.
- Collect real payment data: the bank should record payments so this analysis can use real numbers instead of estimates.